MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP HONOURS , REGULAR EXAMINATION, MARCH 2025

SECOND SEMESTER

MULTIDISCIPLINARY COURSES (MDC)- MG2MDCFSQ100-LIFESTYLE DISEASES AND SOCIAL HEALTH PROBLEMS

DURATION 1.5.hrs MAX.MARKS : 35

SECTION A

MULTIPLE CHOICE QUESTIONS. ANSWER ALL QUESTIONS

Each Question Carries 1 mark)

1.The type of diabetes diagnosed during pregnancy (K , CO1)(1 mark)

c. Gestational Diabetes

2.----------dietary approach is recommened for managing diabetes mellitus (U,CO4)(1 mark)

b. Foods with low fibre

3.Identify a health condition caused by smoking tobacco (K, CO2)(1 mark)

b. Lung cancer

4.Select the reason why underage drinking is particularly harmful(U, CO2)(1 mark))

b. It increases the risk of lifelong addiction

5. Choose the foods that should be avoided for preventing hypertension (K. CO1)(1 mark)

c. Processed meats

6.The systolic blood pressure measures (K,CO1) (1 mark)

b. The pressure when the heart contracts and pumps blood

7. The recommended daily salt intake for individuals with hypertension(K, CO1, CO4)(0.5,0.5mks)

a. Less than 5 g

8. Select the disease that is most commonly caused by smoking (K,CO2)(1 mark)

b. Lung cancer

9. Obesity can be controlled by (U, CO3)(1 mark)

d. Both A and B

10. _____ethnic group is at a high risk of developing Type 2(K, CO4)(1 mark)

b. South Asians

11. Identify which among the following method can be used for preventing STD (K,CO2)(1 mark)

c. Condoms

12.Select a common consequence of smoking at young ones (K,CO2) (1 mark)

a. Incresased risk of lung disease

13. The organ primarily affected by alcoholism, leading tocirrhosis (K,CO2)(1 mark)

c. Liver

14. Lifestyle diseases are dangerous are (K,CO1)(1 mark)

c. Non communicable

15. Obesity is due to (U,CO3)(1 mark)

b. Overnutrition

PART B

SHORT ESSAY QUESTIONS.ANSWER ANY 4 QUESTIONS.

Each question carries 5 marks

16. Illustrate how obesity affects the reproductive health (K,CO3)(5 marks)

Obesity significantly impacts reproductive health in both men and women, leading to various challenges in fertility, pregnancy, and overall health. In women, it can disrupt ovulation, increase the risk of miscarriage, and elevate the chances of pregnancy complications. In men, obesity can impair sperm quality and reduce fertility.

How Obesity Affects Reproductive Health:

In Women:

Disrupted Ovulation:

Obesity can cause irregular periods, amenorrhea (absence of menstruation), and difficulty with ovulation, making it harder to conceive naturally.

Increased Risk of Miscarriage:

Obese women have a higher risk of miscarriage compared to women with a healthy weight.

Pregnancy Complications:

Obesity increases the risk of pregnancy complications like gestational diabetes, high blood pressure (preeclampsia), and the need for cesarean delivery.

Decreased Fertility:

Obesity can lead to reduced fecundity (ability to conceive) even in women who ovulate regularly.

Hormonal Imbalances:

Obesity can disrupt hormone levels, potentially leading to conditions like Polycystic Ovary Syndrome (PCOS), which further impacts fertility.

Reduced Success of Fertility Treatments:

Obese women may experience lower success rates with fertility treatments like IVF due to factors like fewer oocytes and impaired embryo quality.

Increased Risk of Neonatal Complications:

Babies born to obese mothers have a higher risk of birth defects, such as neural tube defects, and other health issues.

Challenges with Diagnosis:

In obese women, it can be more difficult to diagnose fetal abnormalities on ultrasound due to increased body fat.

In Men:

Reduced Sperm Quality:

Obesity can negatively affect sperm count, motility (movement), and morphology (shape), impacting male fertility.

Erectile Dysfunction:

Obesity can contribute to erectile dysfunction, further impacting reproductive health.

Disrupted Hormonal Signaling:

Obesity can disrupt the hypothalamic-pituitary-gonadal axis, affecting testosterone levels and impacting male fertility.

17. Explain the classification of drugs based on their origin with examples (K,CO2)(5 marks)

By definition, drugs are chemical substances that affect or alter the physiology when taken into a living system. They can either be natural or synthetic.

When a drug is therapeutically active and is used for the diagnosis, treatment or prevention of a disease, it is called medicine (legal drugs). They target the macromolecules inside the body and generate a biological response. Most of them interrupt the nervous system (especially the brain) for the generation of a proper biological response. However, they can be toxic in higher doses and generally referred to as lethal doses.

Classification of Drugs on the basis of the Pharmacological Effect:

How a drug or medicine affects or influences the cells of an organism is referred to as the pharmacological effect. Different types of drugs have various pharmacological effects on an organism.

For example, an analgesic reduces pain while an anti-inflammatory drug reduces the inflammation of the body. Thus, drugs can be classified based on the pharmacological effect.

Classification of Drugs on the basis of Drug Action:

Different drugs act differently i.e., each drug has its own way of generating a response called drug action.

Drug action is more specified according to how it generates a response. For example, there are lots of medicines to treat hypertension but each type of drug has different drug actions.

All hypertension medicines reduce blood pressure but in a different pathway.

Classification of Drugs on the basis of Chemical Structure:

This is a common classification of drugs. Generally, drugs that have the same drug action and pharmacological effect have a basic skeletal structure and a minute variation in the branching.

This is why some drugs have more potential than others. For example, all sulphonamides have the same skeletal structure.

Classification of Drugs on the basis of Molecular Targets:

Drugs target the macromolecules inside the body to generate a biological response.

Such macromolecules are called target molecules or drug targets. Drugs that have the same mechanism of action will have the same target.

This basis for the classification of drugs is more helpful during clinical trials.

Drug Classifications Based On Chemical Makeup

Alcohol

Alcohol is the most widely abused substance across most of the world, including in the United States. Legal to some extent in all 50 states, alcohol impacts numerous body systems and causes a host of effects in the user. Alcohol creates feelings of euphoria and lowers inhibitions, but it also severely impairs judgment, perception, and reaction times. Alcohol is a central nervous system (CNS) depressant, but it causes the most severe long-term damage to the liver. There are many types of alcohol, including:

Beer

Wine

Liquor

Opioids

Also called opiates, opioids are either derived from the drug opium or from chemicals designed to mimic it. Opioids work by acting on receptors in the brain, sometimes by mimicking the effects of neurotransmitters. This can enable opioids to serve as powerful painkillers, but also can cause feelings of intense pleasure and lead to addiction. Opioid addiction is one of the most serious problems faced by America today. Opioids are some of the most addictive of all known substances, and they are also some of the deadliest. Some of the most well-known opioids include:

Heroin

Fentanyl

Oxycodone

Benzodiazepines

Benzodiazepines, or benzos, are a class of drugs that function by interacting with the neurotransmitter gamma-aminobutyric acid-A (GABA-A). Each benzo interacts with GABA-A differently, which is why each benzo impacts the body and mind differently. Benzos are prescribed to treat a wide variety of psychiatric and sleep conditions, but they are very commonly abused. Benzos are highly addictive and can cause numerous medical and psychiatric problems when not used as intended. Examples of benzos include:

Ativan

Valium

Xanax

Cannabinoids

Cannabinoids are a class of drugs that are chemically similar to tetrahydrocannabinol (THC), the active agent in marijuana. Cannabinoids create feelings of elation, known as a high, but they also negatively impact mental and physical functioning. Cannabinoids are the most widely abused drugs after alcohol, and they are increasingly gaining legal acceptance. Although considered less addictive than other drug classifications, cannabinoids can seriously damage a person’s mental and physical health. Examples of cannabinoids include:

Marijuana

Hashish

Barbiturates

Barbiturates act on the central nervous system by slowing down its functioning. Barbiturates are derivatives of the chemical barbituric acid. Barbiturates were historically popular for the treatment of psychiatric and sleep disorders, and they are still used for anesthesia and treatment of conditions like epilepsy and headaches. Barbiturates are highly addictive, and they also present a very high overdose risk as they cause many body systems to shut down. Examples of barbiturates include:

Amytal

Luminal

Phenobarbital

Start receiving support via phone, video, or live-chat.

Drug Classifications Based On Effect

Depressants

More commonly referred to as “downers,” depressants create feelings of relaxation and tiredness. While many serve legitimate purposes in the fight against mental illness and sleep deprivation, they are very commonly abused because they may also create feelings of euphoria. Depressants are not only some of the most highly addictivedrugs, but they are also some of the most highly dangerous and likely to cause overdose. Examples of depressants include:

Alcohol

Opiates

Barbiturates

Stimulants

Also known as “uppers,” the primary use of stimulants is to increase energy, concentration, and wakefulness. Stimulants are said to provide a “rush.” In the short term, stimulants are believed to increase productivity and performance while producing an excited high of pleasure. In the long term, stimulants are incredibly addictive and have a very high potential for abuse. Examples of stimulants include:

Adderall

Cocaine

Meth

Hallucinogens

Hallucinogens alter the user’s perception of reality. Often this results in auditory and visual hallucinations, a process known as “tripping.” Although hallucinogens are generally less addictive than other drug classifications, their immediate impacts are generally more severe and dangerous. Examples of hallucinogens include:

LSD

Psilocybin Mushrooms

PCP

Inhalants

Inhalants are a vast range of chemicals that are ingested primarily by breathing them in, or huffing. Most inhalants are commonly used materials that are in no way designed to be ingested by humans. While there is incredible variety between inhalants, most produce feelings of a high. Inhalants are less studied than most other drugs. While they tend to be less addictive than many other substances, the use of inhalants is incredibly dangerous and causes many serious health effects. Examples of commonly abused inhalants include:

Whippits

Nail polish remover

Gasoline

18. Write a note on the environmental and lifestyle factors responsible for Type 2 diabetes (U, CO1, CO4)(2.5 mks, 2.5 mks)

Type 2 diabetes is influenced by a complex interplay of environmental and lifestyle factors, with diet, physical activity, and obesity playing a crucial role. Environmental factors like air pollution, noise, and lack of green spaces can also contribute.

Environmental Factors:

Air Pollution:

Exposure to air pollutants, including fine dust and particulate matter, can increase the risk of type 2 diabetes. Air pollution can act as an endocrine disruptor, affecting the pancreas and liver.

Noise:

Residential noise exposure has been linked to increased risk of type 2 diabetes.

Lack of Green Spaces:

Studies suggest a correlation between lower levels of green spaces and increased risk of type 2 diabetes.

Neighborhood Walkability:

Conversely, higher levels of walkability in neighborhoods have been associated with reduced risk.

Environmental Toxins:

Exposure to certain environmental chemicals, like arsenic, pesticides, and dioxins, may also increase the risk.

Lifestyle Factors:

Obesity:

Being overweight or obese is a major risk factor, particularly visceral obesity (fat around the belly).

Physical Inactivity:

Sedentary lifestyles and lack of physical activity contribute to insulin resistance, a key factor in type 2 diabetes.

Diet:

Diets high in refined carbohydrates, saturated fats, and added sugars are linked to an increased risk.

Alcohol Consumption:

Excessive alcohol consumption can also elevate the risk.

Smoking:

Smoking is a known risk factor for type 2 diabetes and other health problems.

Stress and Sleep:

Chronic stress and disrupted sleep patterns can negatively impact blood sugar control and increase diabetes risk.

Interactions:

Environmental and lifestyle factors often interact in complex ways. For example, air pollution can worsen insulin resistance in individuals who are already overweight or have a sedentary lifestyle. Similarly, living in an area with limited access to healthy food options can contribute to unhealthy eating habits and obesity, further increasing the risk.

19. Explain how positive parenting prevents drug use among children (K,CO2)(5 mks)

The World Health Organisation (WHO) defines substance abuse as harmful or hazardous use of psychoactive substances, including alcohol and illicit drugs. Psychoactive substance use can lead to dependence syndrome - a cluster of behavioural, cognitive, and physiological phenomena that develop after repeated substance use and that typically includes a strong desire to take the drug, difficulties in controlling its use, persisting in its use despite harmful consequences, a higher priority given to drug use than to other activities and obligations, increased tolerance, and sometimes a physical withdrawal state.

child's relationship with his or her parents has a deep impact on their decision making. There is a delicate balance between being a positive role model for your child and being a smothering or overprotective parent. Eventually, teens have to make choices for themselves, and some of those might be poor.

However, parents can take certain actions to reduce the likelihood that their children will experiment with drugs. Some of these actions include:

Know Their Friends

Your child's groups of friends play an immense role in their attitudes on drugs. As children make transition in life, move through school and meet new people, they are likely to bounce around different circles. Some of these people may convince them to experiment with alcohol and other drugs. Parents should get to know their children's friends. It's important to know who your child hangs out with and be cautious of harmful influences. If you notice a negative change in your teen, identify the friends they are spending time with. Encourage your child to hang around the people you consider to be positive influences.

Model Good Behaviour

Socialization process and parents have the biggest influence on the child's life. An unfortunate number of parents dissipate this responsibility with careless personal habits. Something as seemingly innocuous as smoking cigarettes or drinking cocktails can set an example for your child's future behaviour. Parents need to be cognizant of their influence, and should do their best to model good behaviours while in the presence of their children. Research shows that adolescents tend to imitate their parents' behaviour, including alcohol or other drug use.

Create a Positive Environment

Parents need to go beyond modeling good behaviour and should create a positive environment for their child. In teenage years, children need a home that reinforces their good behaviour and builds their self-esteem. Teens who doubt themselves or feel disregarded by their parents are often the ones who seek alcohol or other drugs. Parents should be as involved in their child's life as they can. Motivate your children to strive for excellence and ensure that they have a positive self-image. Diminish stress and practice stress-relief strategies with your children to teach them healthy ways to relax.

Talk to Teens

Early Research studies indicate that approximately nine per cent of parents do not teach their children about the dangers of drug abuse. Overwhelming evidence shows that a parent's lessons and involvement reduce the risk of substance abuse habits, particularly when started at an early age. Therefore, it is crucial for parents to talk to their children early and often about the drugs they may encounter. Let them know the dangers of getting involved with drugs and that you find it unacceptable.

Warn About Peer Pressure

Whether from new friends or strangers, teens will encounter peer pressure during their middle or high school years. When you discuss drug abuse with your kids, be sure to warn them about peer pressure and how to handle it. Explain to children that peer pressure is the influence you feel from others to do something you otherwise would not. A peer could be a friend, coworker, classmate, acquaintance or anyone you admire. Peer pressure may occur directly or indirectly. Direct pressure involves peers explicitly asking you to do something. Indirect pressure happens when you witness others engage in an

activity and are motivated to do the same. The famous slogan “Just Say No” is a basic template for dealing with peer pressure.

Warn children that peer pressure can take many forms, and sometimes the friends your teen trusts the most end up being the ones who encourage them to experiment with drugs. Teach children how to identify these situations, and how to be above the influence of their peers. They may tend to think that saying ‘no’ makes them look uncool. It is important to let them know that resisting peer pressure, and not following the crowd, may be the coolest thing they can do.

Recognize what drugs look like, how they are consumed, and how they are stored

For example, marijuana is generally smoked. This is done by rolling marijuana in paper. It can also be cooked in foods. Crack cocaine is smoked in glass pipes or snorted, typically using a straw, rolled dollar bill, or a small spoon. Heroin is injected typically into the arms or legs. Drugs such as marijuana are stored in small plastic bags, foil packets, or film canisters.

Enforce Consequences for Drug Use

Establish clear rules on the unacceptable use of alcohol and other drugs early in a child’s life. If they violate the rules of the house, make sure there are consequences. A lack of repercussions can lead to repeated experimentation and drug abuse. Parents can foster good behaviour by suspending a teen’s privileges or enforcing some other consequences for abusing drugs.

Monitor Their Activities

Parents should try their level best to keep tabs on their child’s schedule and whereabouts without looking intrusive. If they mention any parties or sleepovers, make sure you know and trust the parents or chaperones who are supervising your children. Children should not have any inhibition or problem sharing this information with you. If they hide or are reluctant to tell you, that might indicate a problem. Do not let your children stay out too late or attend any gatherings that seem suspicious to you.

Parents whose children are into substance abuse should be familiar with the typical hiding places for drugs (especially at home). These include but are not limited to: stuffed toy animals, stereo speakers, the base of lamps, books/magazines, CDs, vents, closets, pillows, parent’s room, car trunks, bushes, soda cans, etc. While monitoring their activities this closely may seem invasive; a study revealed that high levels of parental monitoring are associated with low frequencies of substance use.

Cooperation in the community - School-parent cooperation

For a school-based prevention programme to succeed, strong policies (rules and their consequences) must be developed. This includes policies about the use, possession, and sale of drugs (including alcohol and tobacco). If there is no policy, parents can work with teachers, administrators, and others to develop one. Parents can also ensure that drug education is being taught at all levels and that the information given to students is up to date and age-relevant. Access to referral sources or community resources is also important.

Parent-community activities

When parents support community activities that promote healthy decisions and alternatives, children see this as being a good role model. Parents can help to organize events such as alcohol- and drug-free graduations, serve as chaperones, or help the event by soliciting for funds.

Community co-operation plan

Parents can work with others in their community to increase awareness about the local drug abuse problem and the need for research-based prevention programmes.

Educators can work with others in their school and school system to review current programmes, and identify research-based prevention interventions appropriate for students.

Community Leaders can organize a community group to develop a community prevention plan, coordinate resources and activities, and support research-based prevention in all sectors of the community.

20. Define BMI and explain the purpose of BMI(U,CO3) (5 marks)

BMI, or Body Mass Index, is a calculation that uses height and weight to estimate body fat and assess overall health risk. It's a widely used tool to classify individuals as underweight, normal weight, overweight, or obese, offering an approximate gauge of potential health risks associated with weight.

Purpose of BMI:

Assessing Body Fat:

BMI helps determine if an individual is carrying excess body fat, which is a key factor in many health conditions.

Identifying Health Risks:

BMI is used to estimate the risk of developing chronic diseases like heart disease, high blood pressure, type 2 diabetes, and certain cancers, especially as BMI increases.

Monitoring Weight Trends:

Tracking BMI over time can help individuals and healthcare providers monitor weight loss or gain, as well as track the effectiveness of weight management strategies.

Public Health Research:

BMI data is also used in public health research to understand obesity rates and their impact on different populations, as well as to track changes in obesity prevalence over time.

Limitations of BMI:

Not a Direct Measure of Body Fat:

BMI doesn't directly measure body fat percentage and may not accurately reflect body composition in individuals with high muscle mass or older individuals who have lost muscle.

Ignores Fat Distribution:

BMI doesn't account for the distribution of body fat, which is important, as visceral fat (around the abdomen) poses a greater health risk.

May Not Accurately Reflect Health Risks for All Groups:

BMI might overestimate health risks for some racial and ethnic groups (e.g., Asian individuals) and underestimate them for others (e.g., Black individuals).

Not Suitable for Everyone:

BMI is generally considered a useful tool for adults, but it may not be appropriate for children, adolescents, or pregnant individuals.

21.Describe the methods to treat AIDS (K,CO2) (5 marks)

Acquired immunodeficiency syndrome (AIDS), is an ongoing, also called chronic, condition. It's caused by the human immunodeficiency virus, also called HIV. HIV damages the immune system so that the body is less able to fight infection and disease.

Treatment

There's no cure for HIV/AIDS. Once you have the infection, your body can't get rid of it. But there are medicines that can control HIV and prevent complications.

Everyone diagnosed with HIV should take antiretroviral therapy medicines, also called ART. This is true no matter what stage the disease is in or what the complications are.

ART is usually a mix of two or more medicines from several classes. This approach has the best chance of lowering the amount of HIV in the blood. There are many ART options that mix more than one HIV medicine into a single pill, taken once daily.

Each class of medicines blocks the virus in different ways. Treatment involves mixing medicines from different classes to:

Account for medicine resistance, called viral genotype.

Keep from creating new medicine-resistant strains of HIV.

Suppress the virus in the blood as much as possible.